

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk:

Product Description:

Cat No. :

Synonyms

CAS No

Molecular Formula

**ASID 3-KLOROPEROKSIBENZOIK**
**3-Chloroperoxybenzoic acid, 70-75%**

255790000; 255790010; 255790250; 255791000; 255795000

m-Chloroperbenzoic acid; m-CPBA; MCPBA

937-14-4

C7 H5 Cl O3

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory/Industrial use chemical/reagent. Chemical intermediate.

Uses advised against

Food, drug, pesticide or biocidal product use

**Company**

 Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**E-mail address**

Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888

 CHEMTREC Malaysia **1-800-815-308** (Malay)

 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                   |                     |
|-----------------------------------|---------------------|
| Organic peroxides                 | Type D (H242)       |
| Acute oral toxicity               | Category 4 (H302)   |
| Skin Corrosion/Irritation         | Category 1 C (H314) |
| Serious Eye Damage/Eye Irritation | Category 1 (H318)   |
| Skin Sensitization                | Category 1 (H317)   |
| Acute aquatic toxicity            | Category 1 (H400)   |
| Chronic aquatic toxicity          | Category 1 (H410)   |

**Label Elements**


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## Signal Word

## Danger

### Hazard Statements

H242 - Heating may cause a fire  
H302 - Harmful if swallowed  
H314 - Causes severe skin burns and eye damage  
H317 - May cause an allergic skin reaction  
H410 - Very toxic to aquatic life with long lasting effects

### Precautionary Statements

#### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P220 - Keep away from clothing and other combustible materials  
P234 - Keep only in original packaging  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

#### Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P362 + P364 - Take off contaminated clothing and wash it before reuse

#### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P410 - Protect from sunlight  
P420 - Store away from other materials  
P411 + P235 - Store at temperatures not exceeding 35 °C/ 95 °F. Keep cool

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                  | CAS No    | Weight % |
|----------------------------|-----------|----------|
| 3-Chloroperoxybenzoic acid | 937-14-4  | 70-76    |
| Benzoic acid, 3-chloro-    | 535-80-8  | <20      |
| Water                      | 7732-18-5 | >17      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

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|                                           |                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.                                                                                                                                                                          |
| <b>Skin Contact</b>                       | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.                                                                                                                                                                                                          |
| <b>Ingestion</b>                          | Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water. Never give anything by mouth to an unconscious person.                                                                                                                                                                                               |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                             |

## Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. May cause allergic skin reaction. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

## Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

The product causes burns of eyes, skin and mucous membranes. Do not allow run-off from fire-fighting to enter drains or water courses.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrogen chloride gas.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.

### Environmental precautions

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Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

## **Methods and Material for Containment and Cleaning Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

## **Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### **Precautions for Safe Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance.

### **Conditions for Safe Storage, Including any Incompatibilities**

Keep refrigerated. Keep containers tightly closed in a dry, cool and well-ventilated place. Do not store near combustible materials. Corrosives area.

### **Specific End Uses**

Use in laboratories.

## **SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

### **Control Parameters**

### **Exposure Controls**

#### **Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### **Personal protective equipment**

|                                 |                       |
|---------------------------------|-----------------------|
| <b>Eye Protection</b>           | Goggles               |
| <b>Hand Protection</b>          | Protective gloves     |
| <b>Skin and body protection</b> | Long sleeved clothing |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                 |                                                                                                                 |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>   | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators |
| <b>Recommended Filter type:</b> | Particulates filter conforming to EN 143                                                                        |
|                                 | To protect the wearer, respiratory protective equipment must be the correct fit and be used                     |

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and maintained properly  
When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                                    |                               |                                   |
|----------------------------------------------------|-------------------------------|-----------------------------------|
| Appearance                                         | Off-white                     |                                   |
| Physical State                                     | paste Solid                   |                                   |
| Odor                                               | Slight                        |                                   |
| Odor Threshold                                     | No data available             |                                   |
| pH                                                 | 4.5                           | sat.aq.sol                        |
| Melting Point/Range                                | 92 - 94 °C / 197.6 - 201.2 °F |                                   |
| Softening Point                                    | No data available             |                                   |
| Boiling Point/Range                                | 264 °C / 507.2 °F             | Estimated                         |
| Flash Point                                        | Not applicable                | Method - No information available |
| Evaporation Rate                                   | Not applicable                | Solid                             |
| Flammability (solid,gas)                           | No information available      |                                   |
| Explosion Limits                                   | No data available             |                                   |
| Vapor Pressure                                     | 23 hPa @ 20 °C                |                                   |
| Vapor Density                                      | Not applicable                | Solid                             |
| Specific Gravity / Density                         | 0.56                          |                                   |
| Bulk Density                                       | No data available             |                                   |
| Water Solubility                                   | Slightly soluble              |                                   |
| Solubility in other solvents                       | No information available      |                                   |
| Partition Coefficient (n-octanol/water)            |                               |                                   |
| Component                                          | log Pow                       |                                   |
| 3-Chloroperoxybenzoic acid                         | 1.03                          |                                   |
| Benzoic acid, 3-chloro-                            | 2.7                           |                                   |
| Autoignition Temperature                           | No data available             |                                   |
| Decomposition Temperature                          | > 88°C                        |                                   |
| Viscosity                                          | Not applicable                | Solid                             |
| Explosive Properties                               | No information available      |                                   |
| Oxidizing Properties                               | No information available      |                                   |
| Molecular Formula                                  | C7 H5 Cl O3                   |                                   |
| Molecular Weight                                   | 172.57                        |                                   |
| Self-Accelerating Decomposition Temperature (SADT) | 55°C                          |                                   |

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## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

Yes.

### Chemical Stability

Stable under normal conditions. Organic peroxide. Hazardous decomposition may occur.

### Possibility of Hazardous Reactions

#### **Hazardous Polymerization Hazardous Reactions**

Hazardous polymerization does not occur.  
None under normal processing.

### Conditions to Avoid

Incompatible products. Excess heat. Temperatures above 35°C - SADT = 55°C.

### Incompatible Materials

Reducing Agent. Combustible material. Strong bases.

### Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen chloride gas.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### **Product Information**

#### **(a) acute toxicity;**

**Oral**

Category 4

**Dermal**

Based on available data, the classification criteria are not met

**Inhalation**

Based on available data, the classification criteria are not met

#### **Toxicology data for the components**

| Component                  | LD50 Oral        | LD50 Dermal | LC50 Inhalation |
|----------------------------|------------------|-------------|-----------------|
| 3-Chloroperoxybenzoic acid | 1807 mg/kg (Rat) | -           | -               |
| Water                      | -                | -           | -               |

#### **(b) skin corrosion/irritation;**

**Test method**

Category 1 C

**Test species**

OECD 404

**Observational endpoint**

rabbit

Causes severe irritation and or burns

#### **(c) serious eye damage/irritation;**

**Test method**

Category 1

**Test species**

OECD 405

**Observation end point**

rabbit

CAUSES (SEVERE) EYE BURNS

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**(d) respiratory or skin sensitization;**

**Respiratory**  
**Skin**

Based on available data, the classification criteria are not met  
Category 1

No information available

**(e) germ cell mutagenicity;**

Based on available data, the classification criteria are not met

**(f) carcinogenicity;**

Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;**

Based on available data, the classification criteria are not met

**(h) STOT-single exposure;**

Based on available data, the classification criteria are not met

**(i) STOT-repeated exposure;**

**Target Organs**

Based on available data, the classification criteria are not met

None known.

**(j) aspiration hazard;**

Not applicable  
Solid

**Symptoms / effects, both acute and delayed**

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

**Endocrine Disrupting Properties**

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects**

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component                  | Freshwater Fish                                                  | Water Flea | Freshwater Algae                                                                    | Microtox |
|----------------------------|------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------|----------|
| 3-Chloroperoxybenzoic acid | LC50 (96h): 0.45 mg/l<br>(Cyprinus carpio) OECD<br>Guideline 203 |            | EC50 (72h): 0.49 mg/l<br>(Pseudokirchneriella<br>subcapitata) OECD<br>Guideline 201 |          |

**Persistence and degradability**

**Persistence**  
**Degradation in sewage**  
**treatment plant**

May persist, based on information available.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative potential**

Bioaccumulation is unlikely; May have some potential to bioaccumulate

| Component                  | log Pow | Bioconcentration factor (BCF) |
|----------------------------|---------|-------------------------------|
| 3-Chloroperoxybenzoic acid | 1.03    | No data available             |
| Benzoic acid, 3-chloro-    | 2.7     | No data available             |

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|                                        |                                                                                                              |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <u>Mobility in soil</u>                | Spillage unlikely to penetrate soil. . Is not likely mobile in the environment due its low water solubility. |
| <u>Endocrine Disruptor Information</u> | This product does not contain any known or suspected endocrine disruptors                                    |
| <u>Other adverse effects</u>           | No information available                                                                                     |

## SECTION 13: DISPOSAL CONSIDERATIONS

|                                            |                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Waste treatment methods</u>             |                                                                                                                                                                                                                                                                                                                                      |
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations                                                                                                                                                                  |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition                                                                                               |
| <b>Other Information</b>                   | Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Can be landfilled or incinerated, when in compliance with local regulations Do not empty into drains Large amounts will affect pH and harm aquatic organisms Do not let this chemical enter the environment |

## SECTION 14: TRANSPORT INFORMATION

|                                     |                                                                                           |
|-------------------------------------|-------------------------------------------------------------------------------------------|
| <u>IMDG/IMO</u>                     |                                                                                           |
| UN-No                               | UN3106                                                                                    |
| Hazard Class                        | 5.2                                                                                       |
| Proper Shipping Name                | ORGANIC PEROXIDE TYPE D, SOLID (3-CHLOROPEROXYBENZOIC ACID)<br>3-Chloroperoxybenzoic acid |
| <u>Road and Rail Transport</u>      |                                                                                           |
| UN-No                               | UN3106                                                                                    |
| Hazard Class                        | 5.2                                                                                       |
| Proper Shipping Name                | ORGANIC PEROXIDE TYPE D, SOLID (3-CHLOROPEROXYBENZOIC ACID)<br>3-Chloroperoxybenzoic acid |
| <u>IATA</u>                         |                                                                                           |
| UN-No                               | UN3106                                                                                    |
| Hazard Class                        | 5.2                                                                                       |
| Proper Shipping Name                | ORGANIC PEROXIDE TYPE D, SOLID (3-Chloroperoxybenzoic acid)<br>3-Chloroperoxybenzoic acid |
| <b>Special Precautions for User</b> | No special precautions required                                                           |

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                           |            |
|---------------------------|------------|
| International Inventories | X = listed |
|---------------------------|------------|



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| Component                  | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|----------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| 3-Chloroperoxybenzoic acid | 213-322-3 | X    | X   | X     | X    | X    | -     | X    | KE-05526 |
| Benzoic acid, 3-chloro-    | 208-618-4 | X    | X   | X     | X    | X    | X     | X    | -        |
| Water                      | 231-791-2 | X    | X   | X     | X    |      | X     | X    | KE-35400 |

## National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date**

22-Mar-2025

**Revision Summary**

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

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**End of Safety Data Sheet**